==
ADCS FAULT DETECTION, ISOLATION AND RECOVERY (FDIR)
PROCEDURES MANUAL
ESA OPS-SAT Ground Operations ? Corrective Action Handbook
Version 1.0
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1. FDIR OVERVIEW AND PHILOSOPHY
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Fault Detection, Isolation and Recovery (FDIR) is the structured process
by which ground operators and on-board software detect that an anomaly has
occurred, determine its origin (isolate), and execute actions to restore
nominal operation (recover).

For OPS-SAT, FDIR is particularly important because the mission is designed
to tolerate buggy experimental software. The FDIR hierarchy has three levels:

LEVEL 1 ? ON-BOARD AUTONOMOUS FDIR:
  The ADCS flight software detects anomalies using built-in limit checks
  and sensor consistency monitors. If triggered, it autonomously commands
  safe-mode transitions, sensor resets, or control mode changes. No ground
  contact required. Response time: seconds.

LEVEL 2 ? GROUND-ASSISTED FDIR:
  Anomaly is detected in stored telemetry during a ground station pass.
  Operators review the telemetry, diagnose the cause, and uplink corrective
  commands during the same or next contact window. Response time: minutes
  to ~90 minutes (one orbit).

LEVEL 3 ? ENGINEERING INVESTIGATION:
  Anomaly persists despite Level 1 and 2 responses, or the root cause is
  unknown. Engineering team performs detailed telemetry analysis across
  multiple orbits and potentially develops and uploads a software patch.
  Response time: hours to days.

2. FDIR HIERARCHY AND SEVERITY CLASSIFICATION
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SEVERITY LEVEL 1 ? INFORMATIONAL:
  Anomaly detected but system is operating nominally. Monitor and log.
  Channels: Slight statistical deviation in one channel, within safe limits.
  Action: No immediate action. Schedule trend monitoring.

SEVERITY LEVEL 2 ? CAUTION:
  Anomaly affects one sensor or subsystem but attitude determination is
  still functional with degraded accuracy.
  Action: Ground review at next pass. May plan corrective uplink.

SEVERITY LEVEL 3 ? WARNING:
  Anomaly affects attitude determination accuracy significantly. Mission
  objectives may be impacted. Operator intervention required.
  Action: Priority review. Schedule corrective telecommand uplink within
  1-2 orbits.

SEVERITY LEVEL 4 ? CRITICAL:
  Attitude determination or control is lost. Satellite may be tumbling
  or unable to maintain power-positive orientation.
  Action: Immediate safe-mode commanding if not already triggered on-board.
  All engineering resources mobilised.

3. STEP-BY-STEP DIAGNOSIS PROCEDURE

STEP 1 ? INITIAL TRIAGE (< 5 minutes)
  a. Note which channels have anomaly flags and the anomaly type.
  b. Check orbital phase: is satellite in eclipse? Near polar region?
  c. Check ADCS control mode at time of anomaly.
  d. Check active experiment payloads at time of anomaly.
  e. Record anomaly start time and duration.

STEP 2 ? CONTEXTUAL CORRELATION (5-15 minutes)
  a. Review EPS telemetry: any power switching events?
  b. Review magnetorquer command log: any active torquing?
  c. Review OBC health log: any software resets, I2C errors?
  d. Check space weather: any geomagnetic storm activity?
  e. Is this the first occurrence or a recurring pattern?

STEP 3 ? SUBSYSTEM ISOLATION
  a. Determine if anomaly is isolated to one channel or systemic.
  b. If one magnetometer axis: likely localised EMI or single sensor fault.
  c. If all three magnetometer axes: likely common-cause ? power, EMI,
     radiation event, or attitude control failure.
  d. If photodiodes: correlate with orbital sun geometry first.
  e. If both magnetometers AND photodiodes: suspect attitude loss (tumbling)
     or a power subsystem event affecting all analog sensors.

STEP 4 ? ROOT CAUSE HYPOTHESIS
  Based on all collected evidence, form a prioritised list of root causes
  from most to least likely. Document reasoning. Check against this manual.

STEP 5 ? CORRECTIVE ACTION SELECTION
  Choose the corrective action appropriate to the root cause and severity.
  Always start with the least invasive action (monitoring, configuration
  change) before escalating to safe mode or hardware resets.

STEP 6 ? UPLINK AND MONITOR
  Uplink corrective telecommands during next available contact window.
  Monitor response in subsequent telemetry passes to confirm recovery.
  Document outcome for lessons-learned database.

4. SPECIFIC CORRECTIVE ACTION PROCEDURES

PROCEDURE CA-M01: MAGNETOMETER RECALIBRATION
Purpose: Correct systematic bias in magnetometer readings
Applicable anomalies: B.1 (Sensor Bias Drift)

Steps:
  1. Calculate actual bias = (measured value - IGRF model prediction)
     for a minimum of 5 consecutive telemetry frames.
  2. Average the bias estimates to get a robust bias correction value.
  3. Prepare telecommand: ADCS_MAG_BIAS_CORRECTION_CMD with parameters:
     - Axis: X / Y / Z (select affected axis)
     - Offset: calculated bias value in ?T
     - Apply: TRUE
  4. Uplink during next contact window.
  5. Verify in subsequent pass that corrected readings match IGRF model.

6. Document updated calibration table in operations log.

PROCEDURE CA-M02: MAGNETOMETER SOFT RESET
Purpose: Clear frozen value or noise anomaly by resetting sensor firmware
Applicable anomalies: B.2 (Stuck Value), B.3 (High Noise)

Steps:
1. Confirm anomaly persists over at least 2 telemetry packets.
2. Verify safe to reset: check satellite is not in active de-tumble or
   critical manoeuvre where continuous magnetometer data is essential.
3. Prepare telecommand: ADCS_MAG_RESET_CMD
   - Axis: specific axis or ALL_AXES
   - Type: SOFT_RESET (restarts acquisition thread, does not power cycle)
4. Uplink and monitor. Wait 2 minutes for sensor to reinitialise.
5. If no improvement: escalate to PROCEDURE CA-M03.

PROCEDURE CA-M03: MAGNETOMETER POWER CYCLE
Purpose: Full reset of magnetometer sensor hardware
Applicable anomalies: B.2 (Stuck Value) ? persistent after CA-M02

Steps:
1. Confirm safe operational window (not during critical manoeuvre).
2. Prepare telecommand: ADCS_MAG_POWER_CYCLE_CMD
   - Duration: 5 seconds off
   - Reinit: AUTOMATIC
3. Uplink. Sensor will be unavailable for approximately 30 seconds.
4. During the outage, attitude determination switches to degraded mode.
5. Monitor recovery: sensor should be providing valid data within 60 sec.
6. If power cycle fails to restore: flag as hardware fault. Use remaining
   axes for two-axis attitude determination. Inform ADCS engineering team.

PROCEDURE CA-M04: SWITCH TO STAR TRACKER PRIMARY
Purpose: Remove dependence on degraded magnetometer for attitude
Applicable anomalies: B.2, B.3 when persistent and unrecoverable

Steps:
1. Verify star tracker is operational (check star tracker status TM).
2. Verify satellite is in sunlit portion of orbit (star tracker may be
   blinded if Sun is in its field of view for some attitudes).
3. Prepare telecommand: ADCS_ATTITUDE_SOURCE_CMD
   - Source: STAR_TRACKER_PRIMARY
   - Fallback: MAGNETOMETER (for star tracker lost lock events)
4. Uplink. Monitor attitude determination quality metric in next pass.
5. Continue magnetorquer control (still requires some magnetometer input
   for torque efficiency; use degraded magnetometer with higher uncertainty).

PROCEDURE CA-P01: PHOTODIODE EXCLUSION
Purpose: Remove failed photodiode from sun vector computation
Applicable anomalies: C.1 (Dark channel, not eclipse), C.2 (Saturation)

Steps:
1. Identify failed channel (CADC0875 through CADC0880).
2. Map channel to physical face (±X, ±Y, ±Z).
3. Prepare telecommand: ADCS_PD_MASK_CMD
   - Channel: specify the failed channel ID
   - Mask: EXCLUDE
4. Uplink. ADCS will recompute sun vector using remaining 5 channels.
5. Verify sun vector quality metric in subsequent pass.
6. If 3 or more channels fail: switch to star tracker primary.

PROCEDURE CA-P02: PHOTODIODE CALIBRATION UPDATE
Purpose: Compensate for gradual degradation
Applicable anomalies: C.4 (Gradual Degradation)

Steps:
  1. Compute degradation ratio: current peak / original factory peak
  2. Update gain table: ADCS_PD_GAIN_UPDATE_CMD with new gain coefficients
  3. Verify updated sun vector accuracy against star tracker cross-check.

PROCEDURE CA-S01: TRANSITION TO SAFE MODE
Purpose: Stabilise satellite when attitude control is lost
Applicable anomalies: Any Level 4 critical anomaly

Steps:
  1. If not already in safe mode, uplink: ADCS_SAFE_MODE_CMD
  2. Safe mode activates:
     - B-dot de-tumble controller (uses magnetometers only)
     - Solar power positive orientation (uses photodiodes for sun search)
     - All experiment payloads powered down
     - Minimal power consumption configuration
  3. Monitor de-tumble progress over 3-5 orbits.
  4. Once rates are below 0.5 deg/s, transition back to normal operations.
  5. Perform full ADCS health check before resuming experiments.

5. KNOWN OPERATIONAL SCENARIOS AND EXPECTED RESPONSES
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SCENARIO 1: ECLIPSE ENTRY
  Expected behaviour: All six photodiode channels (CADC0875?CADC0880) drop
  to near-zero simultaneously over 5-10 seconds. ADCS switches to
  magnetometer-only attitude determination. Magnetometer channels remain
  active. This is NORMAL. No action required.

SCENARIO 2: ECLIPSE EXIT (TERMINATOR CROSSING)
  Expected behaviour: Photodiodes gradually increase as satellite exits
  shadow. Channels on the Sun-facing faces brighten first; opposite face
  channels remain dark. Sequence depends on attitude. NORMAL.
  Concern: If photodiodes do not increase within 2 minutes of predicted
  eclipse exit, either attitude is wrong or photodiodes are failed.

SCENARIO 3: DE-TUMBLE AFTER LAUNCH OR SAFE MODE
  Expected behaviour: All ADCS channels show high-frequency variation
  (satellite tumbling). Magnetometers show large swings ±40 ?T as the
  satellite rotates. Photodiodes flash on and off rapidly. This is EXPECTED
  during detumble. Magnetorquers are active. Magnetometer noise is HIGH
  during torquing ? also EXPECTED.
  Recovery: Rates should decrease monotonically over 2-6 hours of B-dot
  detumble. If rates are not decreasing after 6 hours, investigate.

SCENARIO 4: POLAR PASS
  Expected behaviour: Magnetometer readings reach their peak values
  (~55-60 ?T) as the satellite crosses the magnetic poles. This is the
  strongest magnetic field region of the orbit. ADCS may briefly reduce
  magnetorquer authority. NORMAL.

SCENARIO 5: SOUTH ATLANTIC ANOMALY (SAA) PASS
  Expected behaviour: The SAA is a region of weakened magnetic field and
  elevated trapped particle radiation. During SAA passes, single-event
  upset (SEU) rates increase. Transient bit flips in magnetometer ADC

registers are more likely. Expect occasional single-sample spikes.
NORMAL with monitoring. Log SEU events.

SCENARIO 6: MAGNETORQUER DESATURATION MANOEUVRE
Expected behaviour: When reaction wheels approach speed limits, magnetorquers are used to dump angular momentum. During this manoeuvre, magnetometer readings will be noisy and unreliable. Attitude determination quality will degrade for 2-5 minutes. EXPECTED. No action unless manoeuvre does not complete within the expected window.

6. ESCALATION MATRIX

Anomaly Type	Level	First Response	Escalation Trigger
Magnetometer bias >5 ?T	2	CA-M01	No improvement in 3 orbits
Magnetometer stuck value	3	CA-M02 ? CA-M03	CA-M03 fails
Magnetometer high noise	2	Wait for EMI clear	Persists >30 min after EMI stops
All mag. axes anomalous	4	CA-S01 if tumbling	Immediate engineering contact
Single photodiode dark	2	CA-P01	Verify vs. geometry first
All photodiodes dark	4	Check eclipse pred.	If not eclipse: emergency
Photodiode saturation	2	CA-P01 (exclude)	If ADC issue: engineering
Attitude lost (tumbling)	4	CA-S01	Immediate

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END OF FDIR PROCEDURES MANUAL
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